

CODunit - ①Introduction to number system

- Number system or numeral system is the system of naming or representing numbers.
- Number is a mathematical value that helps to count or measure object.
- Generally, number system is defined as a system of writing to express numbers.
- Number system provides a unique representation of every number and represent the arithmetic and algebraic structure of the figure. Also allow us to operate arithmetic operations like addition, subtraction, multiplication and division.

Value of any digit in a number can be determined by.

- The digit
- Its position in the number
- The base of the number system

Type of number System

- Decimal no. system (Base - 10); [0 - 9]
- Binary no. " (Base - 2); [0, 1]
- Octal no. " (Base - 8); [0 - 7]
- Hexadecimal no. " (Base - 16); [0 - F]

① Decimal Number System $\Rightarrow$

- $\rightarrow$ Decimal Number System also called Hindu-Arabic number system.
- $\rightarrow$ Decimal number system has 10-digits represented by 0, 1, ..., 9.
- $\rightarrow$ The base or radix of decimal number system is 10.
- $\rightarrow$ If 25 is decimal number system, 2 is 10 times more than 5.
- $\rightarrow$ Some examples of decimal number systems are $(12)_{10}$, $(345)_{10}$, $(200)_{10}$, $(313.9)_{10}$

Conversion

Decimal number system to other base $\Rightarrow$

Decimal to binary $\Rightarrow$

$$(26)_{10} \longrightarrow (?)_2$$

2	26	0 (MSB)
2	13	1
2	6	0
2	3	1
	1	

$(11010)_2$

(MSB)

Decimal to octal $\Rightarrow$

$$(65)_{10} \rightarrow (?)_8$$

8	65	1 (MSB)
8	8	0
	1 (LSB)	

(101)₈

Decimal to hexadecimal $\Rightarrow$

$$(127)_{10} \rightarrow (?)_{16}$$

16	127	15
	7	

7 (15)
X
F
(7F)₁₆

Conversion from other base to decimal $\Rightarrow$
Binary to decimal

$$(11011)_2 \rightarrow (?)_{10}$$

$$1 \times 2^0 +$$

1	1	0	1	1
				$\rightarrow 1 \times 2^0 = 1$
				$\rightarrow 1 \times 2^1 = 2$
				$\rightarrow 0 \times 2^2 = 0$
				$\rightarrow 1 \times 2^3 = 8$
				$\rightarrow 1 \times 2^4 = 16$

$$(11011)_2 \rightarrow (27)_{10}$$

~~27~~ 27

② Octal to decimal

$$(121)_8 \rightarrow (?)_{10}$$

$$\begin{array}{rcl}
 121 & & \\
 \swarrow \quad \searrow & & \\
 \begin{array}{l}
 \rightarrow 1 \times 8^0 = 1 \\
 \rightarrow 2 \times 8^1 = 16 \\
 \rightarrow 1 \times 8^2 = 64
 \end{array} & & \\
 \hline
 & & 81
 \end{array}
 \quad (81)_{10}$$

③ ~~Octal to~~

③ Hexadecimal to decimal $\Rightarrow$

$$(12)_{16} \rightarrow (?)_{10}$$

$$\begin{array}{rcl}
 12 & & \\
 \swarrow \quad \searrow & & \\
 \begin{array}{l}
 \rightarrow 2 \times 16^0 = 2 \\
 \rightarrow 1 \times 16^1 = 16
 \end{array} & & \\
 \hline
 & & 18
 \end{array}$$

$$(12)_{16} \rightarrow (18)_{10}$$

② Binary Number System $\Rightarrow$

"Bi" in binary means "two".
Hence drawback of this number system is that we represent a number in only 2 terms i.e. 0 and 1.

Binary to

Conversion from binary to other base

① Binary to decimal $\Rightarrow$

$$(10101)_2 \rightarrow (?)_{10}$$

$$\begin{array}{rcl}
 1 & 0 & 1 & 0 & 1 \\
 \downarrow & & \downarrow & & \downarrow \\
 & & & & 1 \times 2^0 = 1 \\
 & & & & 0 \times 2^1 = 0 \\
 & & & & 1 \times 2^2 = 4 \\
 & & & & 0 \times 2^3 = 0 \\
 & & & & 1 \times 2^4 = 16 \\
 & & & & \hline
 & & & & 21
 \end{array}$$

$$(21)_{10} \text{ Ans}$$

② Binary to octal $\Rightarrow$

$$(01101)_2 \rightarrow (?)_8$$

- $\rightarrow$ Firstly convert binary to decimal.
- $\rightarrow$ Then, we will convert the decimal to octal.

$$(01101)_2 \rightarrow (?)_{10} \rightarrow (?)_8$$

$$\begin{array}{rcl}
 0 & 1 & 1 & 0 & 1 \\
 \downarrow & & \downarrow & & \downarrow \\
 & & & & 1 \times 2^0 = 1 \\
 & & & & 0 \times 2^1 = 0 \\
 & & & & 1 \times 2^2 = 4 \\
 & & & & 1 \times 2^3 = 8 \\
 & & & & 0 \times 2^4 = 0 \\
 & & & & \hline
 & & & & 13
 \end{array}$$

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$$(01101)_2 \rightarrow (13)_{10} \rightarrow (7)_8$$

8	13	5
	1	

$$(15)_8$$

$$(01101)_2 = (15)_8$$

~~convert~~

③ Binary to Hexadecimal $\Rightarrow$

$$(1101010)_2 \rightarrow (2)_{16}$$

$$1101010$$

→ Firstly convert binary to decimal.
→ Then, we will convert decimal to hexadecimal.

1	1	0	1	0	1	0	
							$\rightarrow 0 \times 2^0 = 0$ $\rightarrow 1 \times 2^1 = 2$ $\rightarrow 0 \times 2^2 = 0$ $\rightarrow 1 \times 2^3 = 8$ $\rightarrow 0 \times 2^4 = 0$ $\rightarrow 1 \times 2^5 = 32$ $\rightarrow 1 \times 2^6 = 64$ 106

$$(105)_{10}$$

$$(1101010)_2 \rightarrow (106)_{10} \rightarrow (?)_{16}$$

Now convert decimal to ~~hexa~~ hexadecimal

8	105	2
8	13	5
	1	

16	106	A
	6	

$$(6A)_{16}$$

Conversion from other base to binary

① Decimal to binary $\Rightarrow$

$$(294)_{10} \rightarrow (?)_2$$

2	294	0
2	147	1
2	73	1
2	36	0
2	18	0
2	9	1
2	4	0
2	2	0
	1	

$$(100100110)_2$$

② Octal to binary $\Rightarrow$

$$(41)_8 \rightarrow (-?)_8 \rightarrow (?)_2$$

$\rightarrow$ Firstly we will convert octal to decimal.

$\rightarrow$ After that, we will convert decimal to binary.

$$(41)_8 \rightarrow (?)_{10} \rightarrow (?)_2$$

$$(41)_8 \rightarrow (?)_{10}$$

$$\begin{array}{r} 41 \\ \begin{array}{l} \rightarrow 1 \times 8^0 = 1 \\ \rightarrow 4 \times 8^1 = 32 \end{array} \\ \hline 33 \end{array}$$

$$(33)_{10}$$

Now, we will convert

$$(33)_{10} \rightarrow (?)_2$$

2	33	1
2	16	0
2	8	0
2	4	0
2	2	0
	1	1

$$(100001)_2$$

$$(41)_8 \rightarrow (100001)_2$$

③ Hexadecimal to binary $\Rightarrow$

$$(A2B)_{16} \rightarrow (?)_2$$

$\rightarrow$ Firstly, we will convert hexadecimal to decimal.

$\rightarrow$ Now, we will convert decimal to binary.

$$(A2B)_{16} \rightarrow (?)_{10} \rightarrow (?)_2$$

A 2 B

$$\begin{array}{l} \rightarrow B \times 16^0 = 11 \\ \rightarrow 2 \times 16^1 = 32 \\ \rightarrow A \times 16^2 = 2560 \\ \hline 2603 \end{array}$$

$$(2603)_{10}$$

$$(2603)_{10} \rightarrow (?)_2$$

$$\begin{array}{r|l} 2 & 2603 & 1 \\ \hline 2 & 1301 & 1 \\ \hline 2 & 650 & 0 \\ \hline 2 & 325 & 1 \\ \hline 2 & 162 & 0 \\ \hline 2 & 81 & 1 \\ \hline 2 & 40 & 0 \\ \hline 2 & 20 & 0 \\ \hline 2 & 10 & 0 \\ \hline & 5 & \end{array}$$

$$\begin{array}{r|l} 2 & 5 & 1 \\ \hline 2 & 2 & 0 \\ \hline & 1 & \end{array}$$

$$(101000101011)_2$$

$$(A2B)_{16} \rightarrow (101000101011)_2$$

③ Octal Number System $\Rightarrow$

Other base to octal $\Rightarrow$

- ① Binary to octal } Already done
- ② Decimal to octal }
- ③ Hexadecimal to octal $\Rightarrow$

③ Hexadecimal to octal $\Rightarrow$

$$(1BC)_{16} \rightarrow (?)_8$$

$$1 = 0001$$

$$B = 1011$$

$$C = 1100$$

$\rightarrow$ Firstly, we will convert hexadecimal to binary.

$\rightarrow$ Then, we will convert binary to octal.

$$(1BC)_{16} \rightarrow (?)_2 \rightarrow (?)_8$$

$$(1BC)_{16} \rightarrow (000110111100)_2$$

Now Pairing in 3-3.

$$\begin{array}{ccccccc} 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ \hline \downarrow & & & \downarrow & & & \downarrow & & & \downarrow & \\ 0 & & & 6 & & & 7 & & & 4 & \end{array}$$

$$(1BC)_{16} \rightarrow (0674)_8$$

or

$$(1BC)_{16} \rightarrow (674)_8 \quad \underline{\text{Ans}}$$

Octal to other base $\Rightarrow$

- ① Octal to binary } Already done
- ② Octal to decimal }
- ③ Octal to hexadecimal $\Rightarrow$

$\rightarrow$ Firstly, we will convert octal to binary.

$\rightarrow$ Then, we will convert binary to hexadecimal

or

convert octal to decimal. Then decimal to ~~binary~~ hexadecimal.

$$\text{Convert } (56)_8 \rightarrow (?)_{16}$$

$$\cancel{56} (56)_8 \rightarrow (?)_2 \rightarrow (?)_{16}$$

$$(56)_8 \rightarrow (?)_2$$

$$5 = 101$$

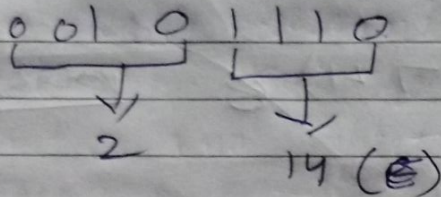
$$6 = 110$$

$$(56)_8 \rightarrow (101110)_2$$

$$(101110)_2 \rightarrow (?)_{16}$$

Now do pairing.

$$(101110)_2 \rightarrow (?)_{16}$$



$$(101110)_2 \rightarrow (2E)_{16}$$

④ Hexadecimal Number System $\Rightarrow$

Other base to hexadecimal $\Rightarrow$

- ① Binary to hexadecimal
 - ② Octal to hexadecimal
 - ③ Decimal to hexadecimal
- Already done

Hexadecimal to other base $\Rightarrow$

① Hexadecimal to ^{decimal} binary $\Rightarrow$

$$\text{Convert } (7CF)_{16} \rightarrow (?)_{10}$$

~~St~~ In hexadecimal

$$7 = 7$$

$$C = 12$$

$$F = 15$$

7CF

$$\begin{aligned} &\rightarrow F \times 16^0 = F = 15 \\ &\rightarrow C \times 16^1 = 12 \times 16 = 192 \\ &\rightarrow 7 \times 16^2 = 7 \times 256 = 1792 \end{aligned}$$

$$\underline{1999}$$

② Hexadecimal to binary $\Rightarrow$

$$(E9A)_{16} \rightarrow (?)_2$$

$$(E9A)_{16} = (1110\ 1001\ 1010)_2$$

③ Hexadecimal to octal $\Rightarrow$

Firstly, we will convert hexadecimal to decimal then decimal to octal

$$(1BC)_{16} \rightarrow (?)_8$$

$$(1BC)_{16} \rightarrow (?)_2 \rightarrow (?)_8$$

$$1 = 0001$$

$$B = 1011$$

$$C = 1100$$

$$(1BC)_{16} = (0001\ 1011\ 1100)_2$$

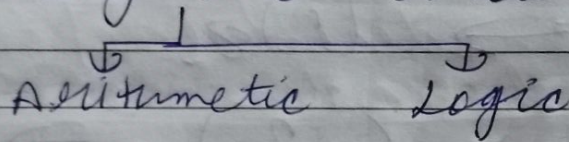
Now, pairing in 3-3.

$$\begin{array}{cccc} 000 & 110 & 111 & 100 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 0 & 6 & 7 & 4 \end{array}$$

$$(1BC)_{16} = (0674)_8 \text{ Ans}$$

Basic organization of computer

- The basic organization of a computer system is the processing unit, memory unit, and input-output devices. The process unit controls all the functions of the computer system. It is
- It is the brain of the computer e.g. CPU
- The memory unit consists of 2 units



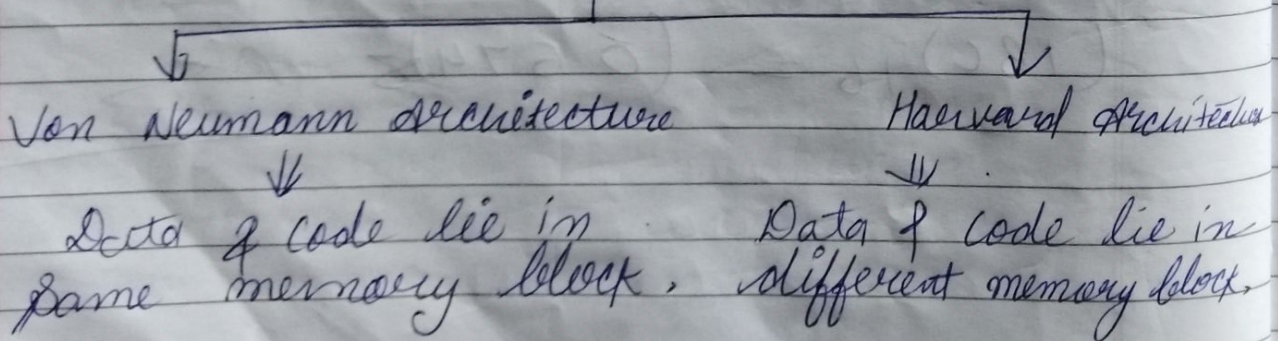
- Input devices
- O/p devices

Von Neumann architecture of computer

Two type of digital computer architecture

- ① Von Neumann architecture
- ② Harvard architecture

In Embedded system type of architecture

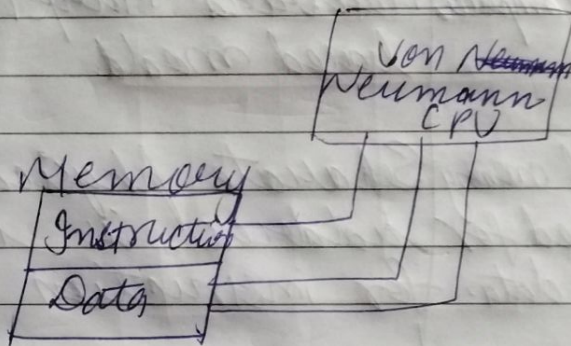


Von Neumann

1 data bus used for both instruction and data.

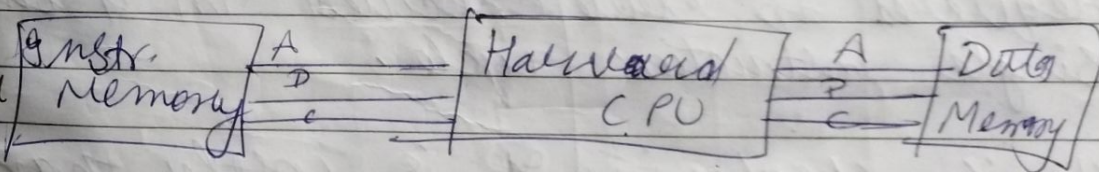
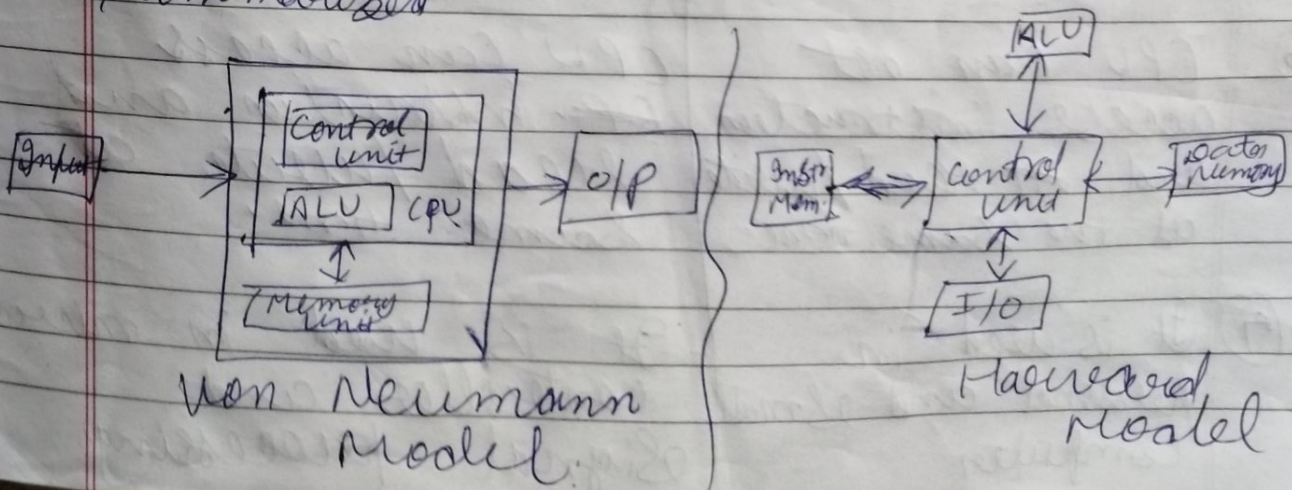


CPU can perform only one operation at a time.



Harvard Architecture :- Separate buses for both instruction & data. Hence in this architecture has data storage CPU entirely contained within the CPU.

This architecture has data storage CPU entirely contained within the CPU. Perform both operation at a same time

Summarized

Difference between Von Neumann and Harvard architecture

S. No.	Von Neumann	Harvard
①	It is an ancient computer architecture based on stored program computer concept.	It is modern computer architecture based on Harvard Mark I relay based model.
②	Same physical memory address is used for instructions and data.	Separate physical memory address is used for instructions and data.
③	There is common bus for data and instruction transfer.	Separate buses are used for transferring data and instruction.
④	2 clock cycle are required to execute single instruction.	An instruction is executed in a single cycle.
⑤	It is cheaper in cost.	It is costly than von Neumann architecture.
⑥	CPU can not access instruction and it read/write at the same time.	CPU can access CP instruction and read/write at same time.
⑦	It is used in personal and small computer.	It is used in microcontroller and signal processing.

Logic Gates $\Rightarrow$ Logic gate is a device that act as a building block for digital ckt. They perform basic logical functions that are fundamental to digital ckt.

I Basic 3 type of logic gates

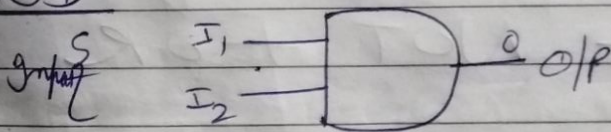
AND

OR

NOT

①

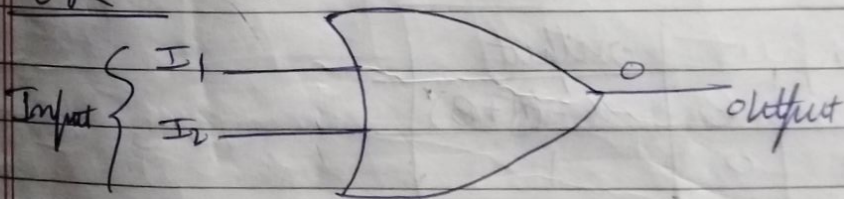
AND



Input		Output
I_1	I_2	O/P
0	0	0
0	1	0
1	0	0
1	1	1

②

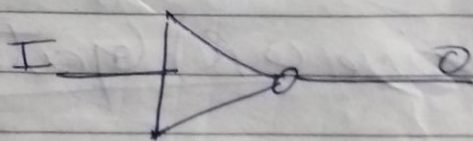
OR



Input		output
I_1	I_2	O/P
0	0	0
0	1	1
1	0	1
1	1	1

③

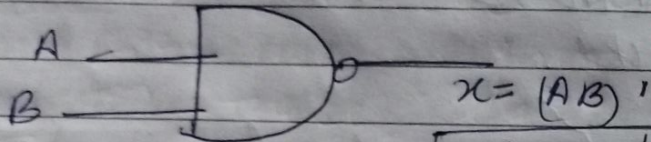
NOT $\Rightarrow$ Also called logical inverters.



Input	output
0	1
1	0

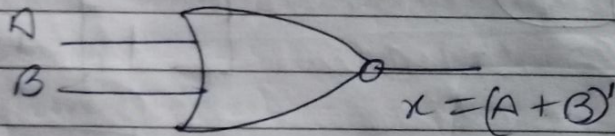
II Universal logic gates

① NAND Gate $\rightarrow$ NOT + AND



Input		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

② NOR Gate $\rightarrow$ NOT + OR



Input		Output
A	B	$X = (A+B)'$
0	0	1
0	1	0
1	0	0
1	1	0

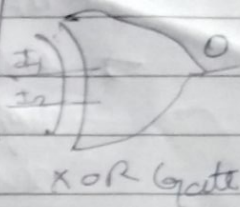
③ XOR (Exclusive OR) Gate

$$X = A \oplus B$$

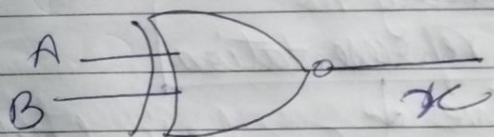
or

$$X = A'B + AB'$$

Input				Output
A	B	A'	B'	$X = (A'B + AB')$
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0



④ Exclusive NOR (XNOR) Gate $\Rightarrow$



$$X = (A \oplus B)'$$

OR

~~$$X = (A'B + AB')$$~~

$$X = A'B' + AB$$

Input		Output	
A	B	A'B'	$X = A'B' + AB$
0	0	1	1
0	1	1	0
1	0	0	0
1	1	0	1

Information Representation $\Rightarrow$

Complements

① 1's

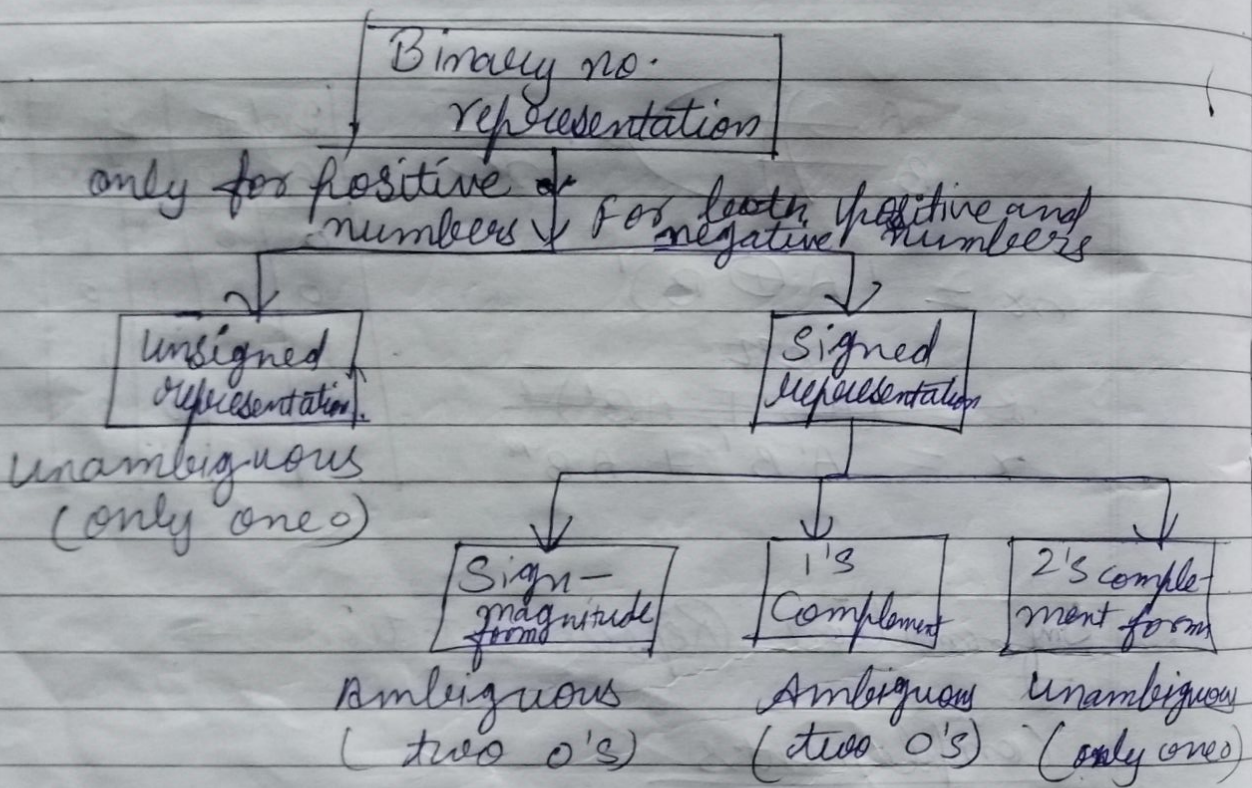
② 2's

Signed and unsigned binary no. system

Representation of binary numbers $\Rightarrow$

Binary number can be represented in signed and unsigned way.

$\rightarrow$ unsigned binary do not have sign bit, whereas signed binary number uses signed bit as well as or these can be distinguishable between positive and negative numbers.



① Unsigned numbers $\Rightarrow$ These no. don't have any sign and can contain only magnitude of the numbers. So ~~it~~ representation of unsigned binary number having all positive numbers.

E.g. Represent 92 in unsigned binary number.

$$(92)_{10} \rightarrow (1011100)_2$$

2	92	0
2	46	0
2	23	1
2	11	1
2	5	1
2	2	0
	1	

1011100

② Signed Number $\rightarrow$ It obtain sign flag, this representation distinguishes positive and negative numbers.

$\rightarrow$ This contains both sign bit and magnitude of a number.

$\rightarrow$ Suppose we want to represent negative decimal numbers, we need to put negative symbol in front of given decimal.

(a) Sign Magnitude form $\rightarrow$ For n bit binary, 1 bit is reserved for sign symbol.

$\rightarrow$ If value of sign bit is 0, given number will be positive.

$\rightarrow$ If value of sign bit is 1, given number will be negative.

Range of sign magnitude is

$$-(2^{(n-1)} - 1) \text{ to } (2^{(n-1)} - 1)$$

(b) 1's complement $\Rightarrow$ 1's complement of a number is given by inverting each bit of a given number.

$\rightarrow$ So we can represent positive numbers in binary form ~~and~~ and negative numbers in 1's complement form.

$\rightarrow$ For sign representation we have

$\rightarrow$ If value of sign bit is 0, number is positive and we can directly represent it in binary form.

$\rightarrow$ If value of sign bit is 1, number is negative and we take 1's complement of a given number.

We can get negative no. by 1's complement of positive number and positive number by using 1's complement of a given number.

Q Find 1's complement of

110010

Solⁿ

001101

Ans

2's complement $\Rightarrow$ It is obtained by inserting each bit of a given number plus 1 to least significant bit (LSB).

Range of 2's complement is

$$(2^{n-1}) \text{ to } (2^{n-1} - 1)$$

Q Find 2's complement of 01101.

Solⁿ 2's complement = 1's complement + 1

$$\begin{array}{r}
 01101 \quad \text{1's complement} \quad 10010 \\
 \downarrow +1 \\
 10010 \\
 + \quad 1 \\
 \hline
 10011
 \end{array}$$

Binary Subtraction $\Rightarrow$

$$0 - 0 = 0$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

$$0 - 1 = 1 \quad (\text{with borrow of } 1)$$

Eg

$$\begin{array}{r}
 1010 \\
 - 101 \\
 \hline
 0101
 \end{array}$$

18-7

$$\begin{array}{r|l}
 2 & 18 \\
 \hline
 2 & 9 \\
 2 & 4 \\
 2 & 2 \\
 & 1
 \end{array}$$

10010

$$\begin{array}{r|l}
 2 & 7 \\
 \hline
 2 & 3 \\
 & 1
 \end{array}$$

111

$$\begin{array}{r}
 10010 \\
 - 111 \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 1101 \\
 - 0111 \\
 \hline
 0110
 \end{array}$$

$$\begin{array}{r}
 0110 \\
 \times 1011010
 \end{array}$$

②

$$\begin{array}{r}
 0110 \\
 \times 1011010
 \end{array}$$

$$\begin{array}{r}
 111111 \\
 - 111111 \\
 \hline
 0010011
 \end{array}$$

$$\begin{array}{r}
 01101010 \\
 \times 1011010 \\
 - 1111 \\
 \hline
 01011
 \end{array}$$

$$\begin{array}{r}
 10010 \\
 - 1111 \\
 \hline
 1011
 \end{array}$$

↓
1' A c

g'car

Carry
+
Add

No can
x' A.C.

Carry

No. Co-

Scarcity

1010 - 0111
using 1's c

1010

+ 1000

70010

$\Rightarrow 0010$

- 85 cm

1 0 1 1
- ~~1~~ 0 1 1

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1011

9 1 1 1

- 2H₂O

1 (

Subtraction using 1's complement

Q $(110101)_2 - (100101)_2$ [with carry]

↓ 1's complement

~~100101~~

011010

$$\begin{array}{r} 110101 \\ + 011010 \\ \hline 11001111 \end{array}$$

↓ Add it
~~011010~~

$$001111$$

$$+ 1$$

$$\hline 010000 \text{ Ans}$$

Q $(101011)_2 - (111001)_2$ [without carry]

↓ 1's complement

000110

$$\begin{array}{r} 101011 \\ + 000110 \\ \hline \end{array}$$

$$- (111001)$$

↓ 1's complement

$$- (001110)$$

Here no carry or extra bit
Again find 1's complement

$$\begin{array}{r} 111001 \\ 3218421 \\ 101011 \end{array}$$

$$43 - 57$$

$$= -14$$

$101011 + 111001$

↓

10

15

$$\begin{array}{r} 20 = 1 \\ 22 = 4 \\ 24 = 16 \\ 25 = 32 \\ \hline 53 \end{array}$$

$$\begin{array}{r} 39 \\ 1 \\ \hline 37 \end{array}$$

Subtraction using 2's complement

with carry

$$(110101)_2 - (100101)_2$$

↓ 1's complement

$$011010$$

↓ 2's complement
+1

$$\underline{011011}$$

$$\begin{array}{r} 110101 \\ + 011011 \\ \hline 1010000 \end{array}$$

Discard

Gross check

$$110101$$

$$\begin{array}{l} 1 \times 2^0 = 1 \\ 0 \times 2^1 = 0 \\ 1 \times 2^2 = 4 \\ 0 \times 2^3 = 0 \\ 1 \times 2^4 = 16 \\ 1 \times 2^5 = 32 \\ \hline 53 \end{array}$$

$$100101$$

$$\begin{array}{l} 1 \times 2^0 = 1 \\ 0 \times 2^1 = 0 \\ 1 \times 2^2 = 4 \\ 0 \times 2^3 = 0 \\ 0 \times 2^4 = 0 \\ 1 \times 2^5 = 32 \\ \hline 37 \end{array}$$

$$53 - 37 = 16$$

$$010000$$

$$\begin{array}{l} 0 \times 2^0 = 0 \\ 0 \times 2^1 = 0 \\ 0 \times 2^2 = 0 \\ 0 \times 2^3 = 0 \\ 1 \times 2^4 = 16 \\ 0 \times 2^5 = 0 \\ \hline 16 \end{array}$$

without carry

$$(101011)_2 - 111001$$

↓ 1's complement
000110

+1

$$\underline{000111}$$

$$\begin{array}{r} 000100 \\ + 101011 \\ + 000111 \\ \hline 110010 \end{array}$$

→ No carry bit
Hence find 2's complement

$$\underline{110010}$$

$$001101$$

+1

$$\underline{001110}$$

$$(Let -ve) \underline{001110} \Rightarrow 14$$

Gross check

$$101011$$

$$\begin{array}{l} 1 \times 2^0 = 1 \\ 1 \times 2^1 = 2 \\ 0 \times 2^2 = 0 \\ 1 \times 2^3 = 8 \\ 0 \times 2^4 = 0 \\ 1 \times 2^5 = 32 \\ \hline 43 \end{array}$$

$$111001$$

$$\begin{array}{l} 1 \times 2^0 = 1 \\ 0 \times 2^1 = 0 \\ 1 \times 2^2 = 4 \\ 1 \times 2^3 = 8 \\ 1 \times 2^4 = 16 \\ 1 \times 2^5 = 32 \\ \hline 57 \end{array}$$

Binary Addition $\Rightarrow$ It work same as base 10 decimal system, except that it is base 2 system. The binary system consists of 2 digits i.e 0 and 1.

Rules

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 0 \text{ with carry } 1 [10]$$

Procedure for binary addition $\Rightarrow$

$$\begin{array}{r} \text{Q(1)} \quad 101 \\ + 101 \\ \hline 1010 \end{array}$$

$$\begin{array}{r} \text{Q(2)} \quad 1011100010 \\ + 11000 \\ \hline 1001000 \end{array}$$

Binary Addition Using 1's complement

Case ① $\Rightarrow$ when a positive number has a greater magnitude

$$+1111 \quad \text{and} \quad -1101$$

$$+1111 = 0111$$

$$-1101 = 1101 \quad [\text{Taking 1's complement}]$$

$$\downarrow$$

$$00010$$

Hence result is +0010 Ans

15

13

Q. $+1111$ and -1101

Sol. $+1111 = 01111$
 $-1101 = \overline{11101}$
 $\downarrow$ 1's complement
 10010

Add

$$\begin{array}{r} 01111 \\ + 10010 \\ \hline 100001 \\ \quad + \uparrow \\ \quad \downarrow \\ 00001 \\ + 1 \end{array}$$

$$\boxed{01010} \quad \text{Ans}$$

* Case (2) $\rightarrow$ When negative no. has a greater magnitude

Q. -1111 and $+1101$ $(-15) + 13$

$$-1111 = \overline{00001111}$$

$\downarrow$ 1's complement
 10000

 $\downarrow$ Now add

$$\begin{array}{r} 10000 \\ + 01101 \\ \hline 11101 \end{array}$$

 $\downarrow$ 1's complement

$$10010$$

Case ③ $\Rightarrow$ Addition of two negative no.

a -1010 and -0011

$$\begin{aligned} -1010 &= 1\ 1010 \xrightarrow{\text{1's complement}} \{1\ 0101 \\ -0011 &= 1\ 0011 \xrightarrow{\text{1's complement}} \{1\ 1100 \end{aligned}$$

$$\begin{array}{r} 1\ 0101 \\ + 1\ 1100 \\ \hline 1\ 0001 \\ \xrightarrow{+1} \\ 1\ 0010 \end{array}$$

Therefore, taking 1's complement

$$\begin{array}{r} 1\ 1101 \\ \downarrow \\ -1101 \text{ Ans } \end{array}$$

Binary Multiplication $\Rightarrow$

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

$$\begin{array}{r}
 110 \\
 \times 100 \\
 \hline
 000 \\
 000 \times \\
 110 \times \times \\
 \hline
 11000
 \end{array}$$

$$\begin{array}{r}
 1100 \Rightarrow 12 \\
 \times 1111 \Rightarrow 15 \\
 \hline
 1100 \\
 1100 \times \\
 1100 \times \times \\
 1100 \times \times \times \\
 \hline
 10110100 \\
 \hline
 \begin{array}{ccccccc}
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1
 \end{array}
 \end{array}$$

$$128 + 32 + 16 + 4$$

$$180$$

S.No

Fixed point

①

It has fixed no. of bits for integer part and for fractional part.

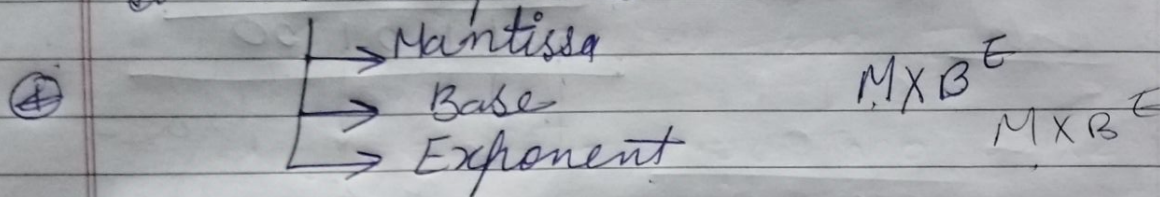
Floating Point

Floating Point offers that a no.'s radix point (decimal point or more commonly in computer, binary point) can float. That is it can be placed anywhere relative to the significant digits of the no.

Floating point representation $\Rightarrow$ Bin

Binary point floats to the right of most significant 1 and an exponent is used.

It has 3 points $\Rightarrow$

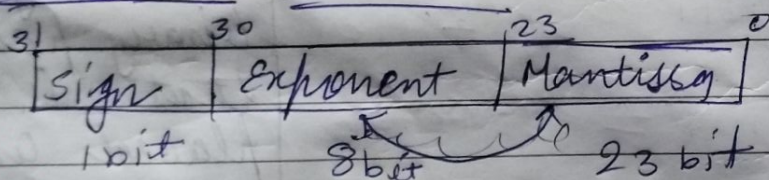


E.g.

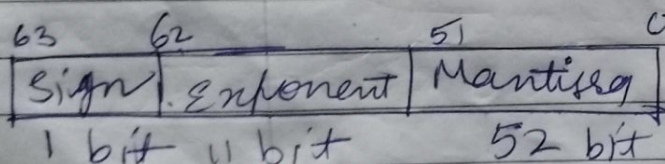
Number	Mantissa	Base	Exponent
① 9×10^8	9	10	8
② 110×2^7	110	2	7
③ 4364.784	4364.784	10	-3

IEEE⁷⁵⁴ floating point representation $\Rightarrow$

① Single Precision Format $\Rightarrow$ 32 bit



② Double Precision Format $\Rightarrow$ 64 bit



Ex 8 Represent $(1259.125)_{10}$ in Single and double precision format.

Solⁿ Step ① convert decimal to binary

$$(1259)_{10} = \cancel{110011001} \\ (0.125)_{10}$$

2	1259	1
2	629	1
2	314	0
2	157	0
2	78	2
2	39	1
2	19	1
2	9	1
2	4	0
2	2	0
	1	1

$$10011100011$$

$$\cancel{10011101001}$$

$$(1259)_{10} = (10011100011)_2$$

$$(0.125)_{10} = (0010)_2$$

$$0.125 \times 2 \Rightarrow 0.250 \Rightarrow 0$$

$$0.250 \times 2 \Rightarrow 0.500 \Rightarrow 0$$

$$0.5 \times 2 \Rightarrow 1.0 \Rightarrow 1$$

$$0.0 \times 2 \Rightarrow 0 \Rightarrow 0$$

$$0010$$

$$(1259.125)_{10} = (10011100011.0010)_2$$

Step ② $\Rightarrow$ Normalize the numbers

Single Precision

$$(1.N) 2^{E-127}$$

Double Precision

$$(1.N) 2^{E-1023}$$

$$2^{E-127}$$

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$E-127=10$

$E=137$

$$(10011100011001)_2$$

$$1.0011100011 \times 2^{10}$$

Step 3 H

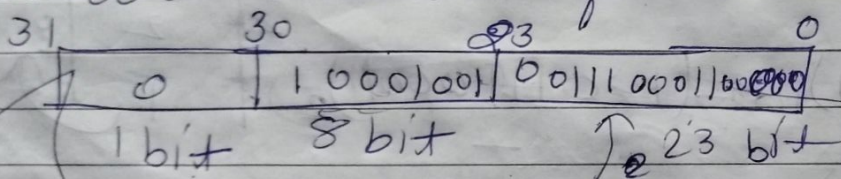
Single Precision format

$$(1.N)_2 E-127 = 1.0011100011001 \times 2^{10}$$

$$E-127=10$$

$$E=137 \Rightarrow (10001001)_2$$

~~Double Precision format~~



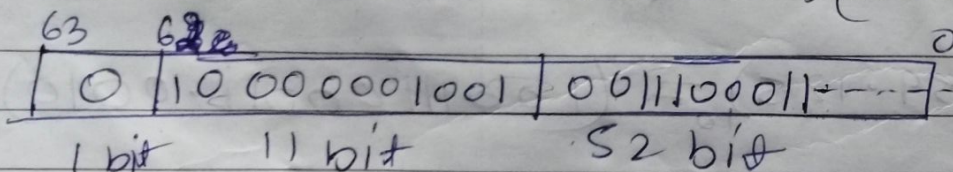
Step 4 Double Precision Format

$$(1.N)_2 E-1023 = 1.0011100011 \times 2^{10}$$

$$E-1023=10$$

$$E=1023+10$$

$$E=1033 \Rightarrow (10000001001)_2$$



Unit - II

K-Map $\rightarrow$ It is also called (Karnaugh Map) developed by ~~albert~~ ^{albert} in 1953, ~~Karnaugh~~ ^{Karnaugh} Maurice Karnaugh.

- $\rightarrow$ K map is a graphical representation used for simplifying the boolean expression.
- $\rightarrow$ For a boolean expression consisting of n -variables, number of cells required in K map = 2^n cells.
- $\rightarrow$ K map is based on Grey code unit (Unit distance code).
- $\rightarrow$ K map is based on 3 type of input values (0, 1, don't care).

	A	B
	0	0
2 bit change	0	1
	1	0
	1	1

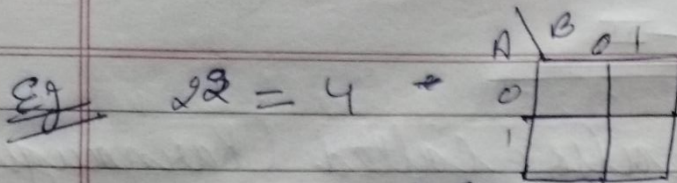
1 bit change

BC 00 01 ~~10~~ 11 10

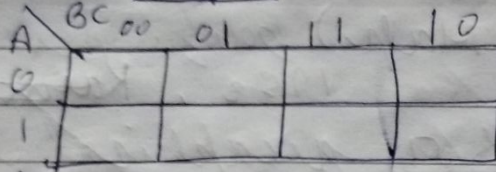
A	0	1	2	3
0	000 0	001 1	011 3	010 2
1	100 4	101 5	111 7	101 6

K map is used to simplify the boolean expression in pictorial format without using boolean laws or theorems.

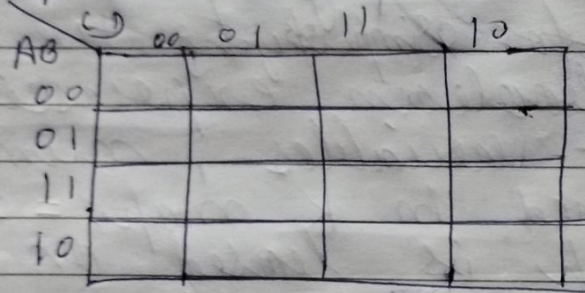
Formula $\rightarrow 2^n$ where n is the no. of variable.



$2^3 = 8$



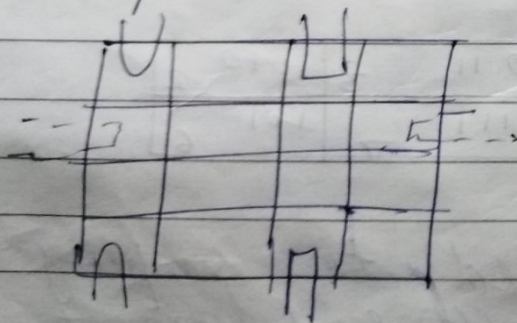
$2^4 = 16$



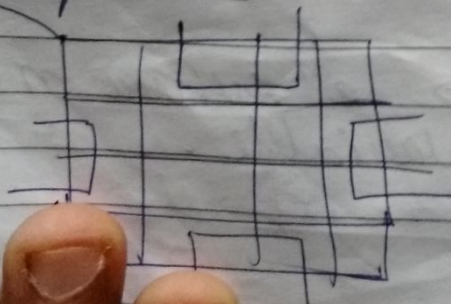
Rules for creating group in K-map $\Rightarrow$

- ① NO zeros allowed.
- ② Group can be horizontal or vertical but can't be diagonal.
- ③ Overlapping allowed
- ④ Group should be as large as possible.
- ⑤ Groups must contain 2^n cells.

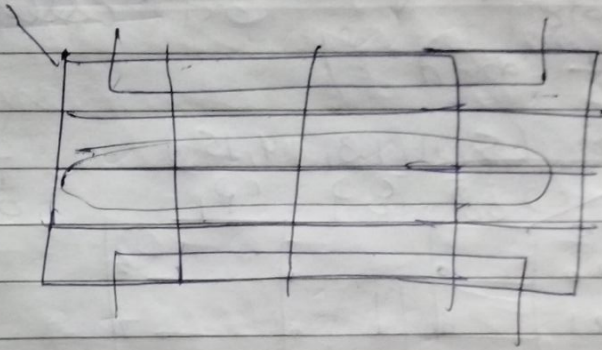
① Pair Groups $= 2$ cells



② Quad Group $= 4$ cells



③ Oct Group $\Rightarrow$ 8 cells



4 variable K map

		CD			
		00	01	11	10
AB	00	0	1	3	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

$F(A, B, C, D)$

$$Q \quad F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

PQ	RS			
	00	01	11	10
00	0	1	X 3	2
01	4	5	7	6
11	X 12	13	15	X 14
10	X 8	9	11	X 10

Essential Prime
implicant

ऐसा group जिसका का से का एक 1
 ऐसा हो जो किसी और group में
 participant नहीं करे उसे essential
 prime implicant कहते हैं

$$F(A, B, C, D) = \sum m(5, 11, 13, 14, 15)$$

Q

	AB	CD 00	01	11	10
$\bar{A}\bar{B}$ 00		0	1	3	2
$\bar{A}B$ 01		4	5	7	6
AB 11		12	13	15	14
$A\bar{B}$ 10		8	9	11	10

$\swarrow$ ABD
 $\swarrow$ BCD
 $\swarrow$ ACD
 $\swarrow$ ABC

Prime implicant = ABD

Essential prime implicant = ACD, ABC, BCD

SOP & POS $\Rightarrow$

SOP (Sum of product)

POS (Product of Sum)

SOP and POS representation for logic expressions $\Rightarrow$

① Sum of product (SOP) form :-
 Group of product terms summed together.

$$Y = AB + AC + BC$$

product term

Sum $\rightarrow$ (or) +

Product $\rightarrow$. (AND)

Eg $y = ABC + BCD + ABD$
 $y = \bar{P}Q + PQR + PQR$

A, B, C, D, P, Q, R

$\hookrightarrow$ i/p terms or literals

Product of Sums (POS) form $\Rightarrow$

$$y = (A+B)(B+C)(A+C)$$

Sum terms

Group of Sum terms multiplied together.

E.g. $y = (A + \bar{B} + \bar{C}) \cdot (A+B) \cdot (A+C)$

$$y = (P+R) \cdot (P+\bar{Q}) + (\bar{P}+R)$$

$\downarrow$ OR $\downarrow$ OR $\downarrow$ OR

Standard / canonical SOP form $\Rightarrow$

In this each product term consists of all the literals in the complemented or uncomplemented form.

eg (A, B, C) are three literals.

$$Y = ABC + A\bar{B}\bar{C} + \bar{A}BC$$

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Standard/canonical POS form $\Rightarrow$ In this each sum term consists of all the literals in the complemented or uncomplemented form.

eg (A, B, C)

$$\text{then } Y = (A+B+\bar{C}) \cdot (A+\bar{B}+C) \cdot (\bar{A}+B+\bar{C})$$

Check whether the following are standard form or not.

① $Y = AB + AB\bar{C} + \bar{A}BC$

$\hookrightarrow$ (SOP) Not in standard/canonical SOP. [Non Standard SOP]

② $Y = AB + A\bar{B} + \bar{A}\bar{B}$

$\hookrightarrow$ (SOP) standard SOP

③ $Y = (\bar{A}+B) (A+B) \cdot (A+\bar{B})$

$\hookrightarrow$ (POS) standard POS

④ $Y = (\bar{A}+B) \cdot (A+B+C)$

$\hookrightarrow$ (POS) Non standard POS

Minterm and Maxterm $\Rightarrow$

Minterm $\Rightarrow$ Each individual term in the standard SOP form is called minterm.

E.g. $Y = ABC + A\bar{B}\bar{C} + \bar{A}BC$

$\hookrightarrow$ Minterm $\sum_{i=0}^{\infty} m_i$

Maxterm $\Rightarrow$ Each individual term in the standard POS form is called maxterm.

$Y = (A+B) (A+\bar{B})$

$\prod_{i=0}^{\infty} M_i$

$$\underline{AB}$$

$$\bar{A} + \bar{B} \quad A \quad B$$

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Q. Write minterms and maxterms for the following truth table.

Standard (SOP)

Standard (POS)

A	B	Minterms (m_i)	Maxterms (M_i)
0	0	$\bar{A} \bar{B} \rightarrow m_0$	$A + B$
0	1	$\bar{A} B \rightarrow m_1$	$A + \bar{B}$
1	0	$A \bar{B} \rightarrow m_2$	$\bar{A} + B$
1	1	$AB \rightarrow m_3$	$\bar{A} + \bar{B}$

How to represent logical expression with minterms and maxterms.

① $Y = ABC + \bar{A}BC + A\bar{B}\bar{C}$

A	B	C	$m_i (\Sigma)$	$M_i \Pi$
0	0	0	$\bar{A}\bar{B}\bar{C} \rightarrow m_0$	$A+B+C \rightarrow m_0$
0	0	1	$\bar{A}\bar{B}C \rightarrow m_1$	$A+B+\bar{C} \rightarrow m_1$
0	1	0	$\bar{A}B\bar{C} \rightarrow m_2$	$A+\bar{B}+C \rightarrow m_2$
0	1	1	$\bar{A}BC \rightarrow m_3$	$A+\bar{B}+\bar{C} \rightarrow m_3$
1	0	0	$A\bar{B}\bar{C} \rightarrow m_4$	$\bar{A}+B+C \rightarrow m_4$
1	0	1	$A\bar{B}C \rightarrow m_5$	$\bar{A}+B+\bar{C} \rightarrow m_5$
1	1	0	$AB\bar{C} \rightarrow m_6$	$\bar{A}+\bar{B}+C \rightarrow m_6$
1	1	1	$ABC \rightarrow m_7$	$\bar{A}+\bar{B}+\bar{C} \rightarrow m_7$

$$Y = \underbrace{ABC}_{m_7} + \underbrace{\bar{A}BC}_{m_3} + \underbrace{A\bar{B}\bar{C}}_{m_4}$$

$$Y = m_7 + m_3 + m_4$$

$$Y = \Sigma m(3, 4, 7)$$

Ans

Q② $Y = (A + \bar{B} + C) (A + B + C) (\bar{A} + \bar{B} + C)$

From table $M_2 \cdot M_0 \cdot M_6$

$$Y = M_2 \cdot M_0 \cdot M_6$$

$$Y = \bar{A} \bar{B} C + A B C + A \bar{B} C$$

$$Y = \sum m(0, 2, 6) \text{ Ans}$$

How to write standard SOP expression for a given Truth Table $\Rightarrow$

- ① Consider only input combination whose output is 1.
- ② Write product term for each such combination.
- ③ 'OR' all these product terms.

eg

A	B	Y	
0	0	0	
0	1	1	$\bar{A} B$
1	0	1	$A \bar{B}$
1	1	0	

$$Y = \bar{A} B + A \bar{B}$$

$$Y = m_1 + m_2$$

$$Y = \sum m(1, 2) \text{ Ans}$$

Eg

A	B	C	Y
0	0	0	0
0	0	1	1 $\rightarrow \bar{A} \bar{B} C$
0	1	0	0
0	1	1	0
1	0	0	1 $\rightarrow A \bar{B} \bar{C}$
1	0	1	0
1	1	0	0
1	1	1	1 $\rightarrow ABC$

$$\underbrace{\bar{A} \bar{B} C}_{m_1} + \underbrace{A \bar{B} \bar{C}}_{m_4} + \underbrace{ABC}_{m_7}$$

$$Y = m_1 + m_4 + m_7$$

$$Y = \sum m(1, 4, 7)$$

How to write Standard POS Expression of given truth table?

- ① Consider only those combinations of i/p which produce low output ($Y=0$).
- ② Write maxterms only for such i/p combination.
- ③ AND - these maxterms.

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

A	B	C	Y
1	1	1	1

$$\rightarrow A + B + C$$

$$\rightarrow \bar{A} + B + \bar{C}$$

$$\rightarrow \bar{A} + \bar{B} + C$$

$$(A+B+C) \cdot (A+\bar{B}+\bar{C}) \cdot (\bar{A}+B+\bar{C}) \cdot (\bar{A}+\bar{B}+C)$$

$$M_0 \cdot M_3 \cdot M_5 \cdot M_6$$

$$Y = \text{TTM} (0, 3, 5, 6)$$

Algebraic Simplification of Boolean Expression $\Rightarrow$

- ① Algebraic Method
- ② K Map method
- ③ Quene

Simplification of boolean expression using K-Map

- ① Use grouping technique
- ② Grouping means combining terms in adjacent cells.
- ③ Grouping adjacent 1's $\Rightarrow$ SOP
- ④ Grouping adjacent 0's $\Rightarrow$ POS

Example of Grouping Pairs $\Rightarrow$

$$Y = \bar{A}BC + \bar{A}\bar{B}\bar{C}$$

	BC	$\bar{B}\bar{C}$	$\bar{B}C$	$B\bar{C}$
$\bar{A}$	0	0	1	1
A	0	0	0	0

In groups look for variables whose value is not changing
 $\bar{A} B$ Ans

eg 2) $Y = \bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}D$

AB \ CD	00	01	11	10
00	0	1	0	0
01	0	0	0	0
11	0	0	0	0
10	0	1	0	0

$\bar{B}\bar{C}D$ Ans

Q $Y = \sum m(1, 2, 3, 4, 5, 7, 9, 11, 13, 15)$

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

Pair 1, $\bar{A}\bar{B}C$

Octad D

Pair 2 $\bar{A}BC$

$Y = \bar{A}BC + D + \bar{A}\bar{B}C$ Ans

8) $F(A, B, C, D) = \prod M(0, 2, 6, 7, 8, 10, 12, 13)$

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	1	1	0
11	0	0	1	1
10	0	1	1	0

quad 1 $\Rightarrow B$

(B + D)

Pair 1 $\Rightarrow (A + \bar{B} + \bar{C})$

Pair 2 $\Rightarrow (\bar{A} + \bar{B} + C)$

$Y = (B + D) \cdot (A + \bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B} + C)$

5 variable K Map का वही है वही?

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Logic Design

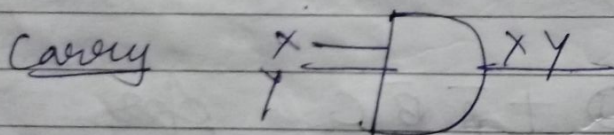
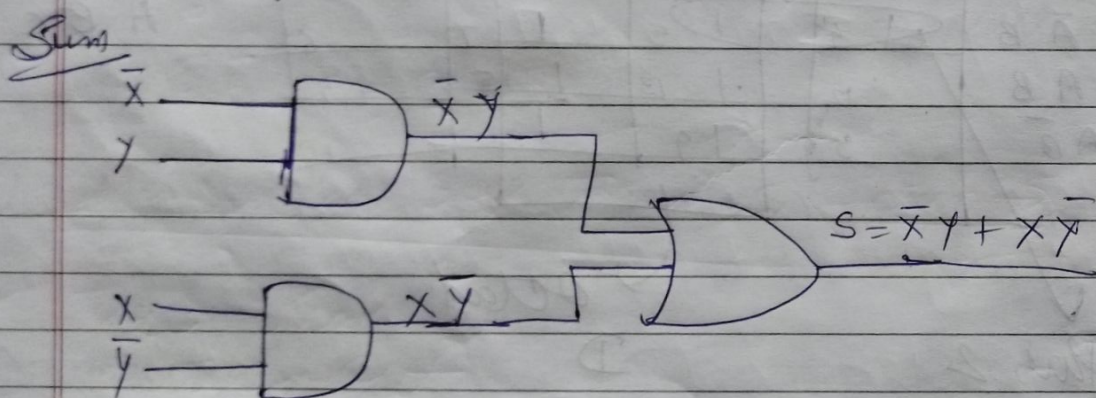
① Half Adder $\Rightarrow$ Also called 2 input and 2 output circuit

X	Y	(MSB) Carry	(LSB) Sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

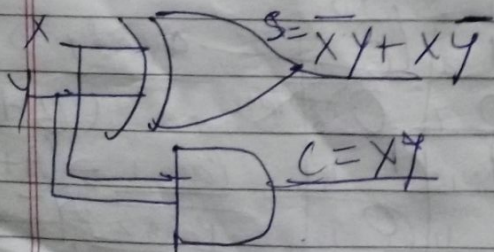
Minterms

$$S \Rightarrow \bar{x}y + x\bar{y}$$

$$C \Rightarrow xy$$



But here $S = \bar{x}y + x\bar{y}$ means XOR Gate



Full Adder

X	Y	Z	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

0 0 0

1 1 1

1 1 0 1

1 0 1 0 0

Minterms

$$\text{Sum} = (\bar{x}\bar{y}z) + (\bar{x}y\bar{z}) + (x\bar{y}\bar{z}) + (xyz)$$

$$\text{Carry} = \bar{x}yz + x\bar{y}z + x\bar{y}\bar{z} + xyz$$

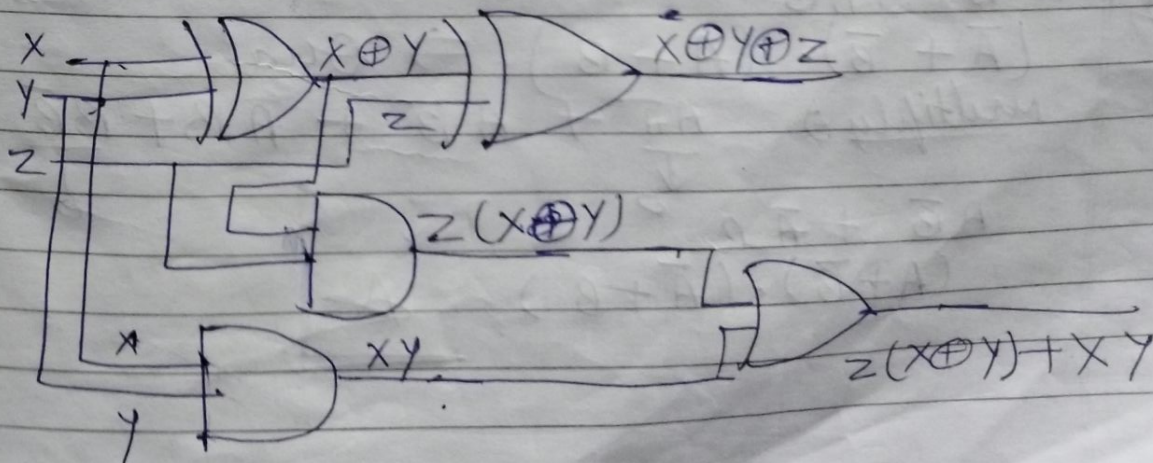
$$\text{Sum} = x \oplus y \oplus z$$

~~$$\text{Carry} = xy + yz + zx$$~~

$$\begin{aligned} \text{Carry} &= \bar{x}yz + x\bar{y}z + x\bar{y}\bar{z} + xyz \\ &= z(\bar{x}y + x\bar{y}) + xy(z + \bar{z}) \end{aligned}$$

↓

$$\text{Carry} = z(x \oplus y) + xy$$



Solider Subtractor

SOP & POS conversion

Conversion between SOP and POS form

① POS to SOP form:-

- Multiply out
- Remove redundancy

Eg $(A + \bar{B})(\bar{A} + B)$

$$\begin{array}{ccccccc} A & \bar{A} & + & \bar{A} & \bar{B} & + & A & B & + & B & \bar{B} \\ \downarrow & & & & & & & & & \downarrow & \\ 0 & & & & & & & & & 0 & \\ \bar{A} & \bar{B} & + & A & B & & & & & & \end{array}$$

② Conversion SOP to POS form:-

- Take dual
- Multiply out
- Remove redundancy
- Take dual.

Eg $\bar{A} \bar{B} + AB$

$(\bar{A} + \bar{B}) \cdot (A + B) \Rightarrow \text{Dual}$

Multiply $\Rightarrow \begin{array}{ccccccc} A & \bar{A} & + & A & \bar{B} & + & \bar{A} & B & + & B & \bar{B} \\ \downarrow & & & & & & & & & \downarrow & \\ 0 & & & & & & & & & 0 & \end{array}$

$A \bar{B} + \bar{A} B$

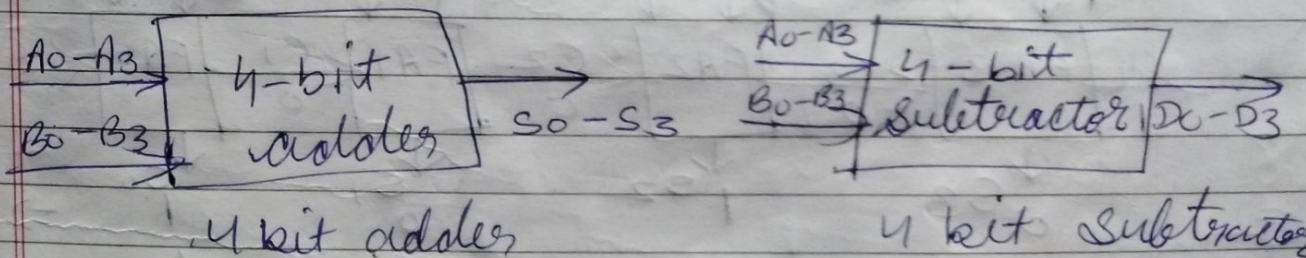
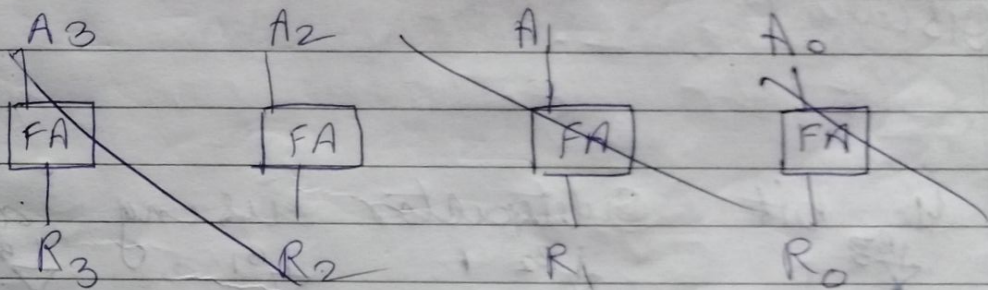
$(A + \bar{B}) \cdot (\bar{A} + B)$

A

Adder Subtractor $\Rightarrow$ This circuit is capable for adding or subtracting numbers.

- $\rightarrow$ In this ckt adds or subtracts depending on a control signal.
- $\rightarrow$ It is also possible to construct a ckt that performs both addition and subtraction at the same time.

$$\begin{array}{r} A \\ + B \\ \hline \end{array}$$



Subtraction using adder

$$\begin{array}{r} 14 \\ - 5 \\ \hline \end{array} \qquad \begin{array}{r} 1110 \\ - 0101 \\ \hline \end{array}$$

$$14 + (2^4 \text{ complement of } 5)$$

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2's complement of 5

$$0101 \xrightarrow{1's} 1010 \xrightarrow{+1} = 1011$$

$$14 + (1011)$$

$$1110 + 1011$$

$$00$$

$$1110$$

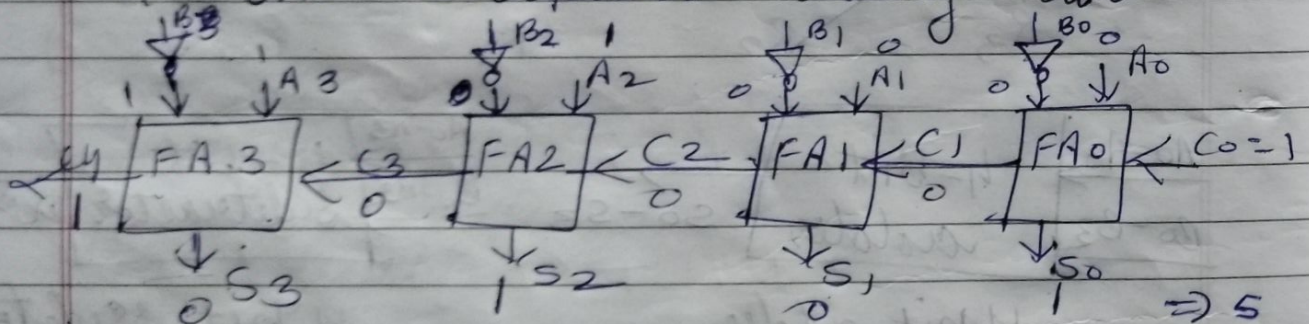
$$+ 1011$$

$$\begin{array}{r} 1001 \\ \hline \end{array}$$

Discard it

$$1001 \rightarrow 9$$

4 bit Subtractor using adder.



$$A_3 A_2 A_1 A_0 + 2^3 \text{ complement of } (B_3, B_2, B_1, B_0) \Rightarrow 5$$

$$\begin{array}{r} 12 \\ - 7 \\ \hline \end{array}$$

$$\begin{array}{r} 1100 \\ - 0111 \\ \hline \end{array}$$

12 + (2's complement of 7)

$$1100 + 1001$$

$$\begin{array}{r} 01100 \\ + 1001 \\ \hline \end{array}$$

$$\begin{array}{r} 01100 \\ + 1001 \\ \hline 10101 \end{array}$$

Discard

$$0101 \Rightarrow 5$$

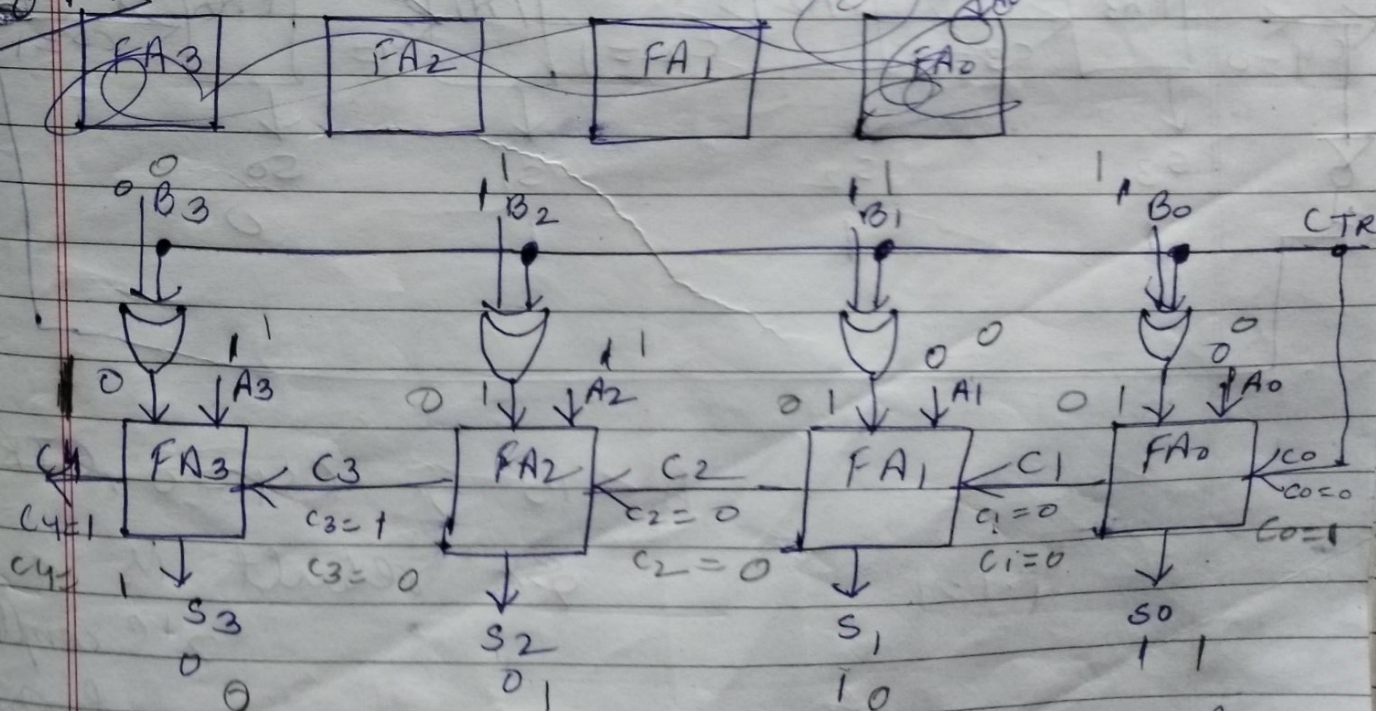
4 bit adder / subtractor $\Rightarrow$

$$A \oplus 0 = A \cdot 0 + \bar{A} \cdot 0 = A \cdot 0 = 0$$

$$A \oplus 1 = A \cdot 1 + \bar{A} \cdot 0 = A \cdot 1 = A$$

$$A \oplus A = A \cdot A + \bar{A} \cdot \bar{A} = A \cdot A + \bar{A} \cdot \bar{A} = A \oplus A = 0$$

$$\begin{array}{r} 1100 \\ - 0111 \\ \hline 1001 \end{array}$$



$12 + 7 = 10011$ (A > B)
 $12 - 7 = 0101$ (A < B)